

INT202 Complexity of Algorithms

1. In the RSA method, suppose that $p = 5$, $q = 17$, and $e = 13$. First find the private key d corresponding to these parameters. Then decrypt the ciphertext messages, C , below to find the original (plaintext) messages.

a. $C = 12$

b. $C = 9$

Solution:

Here we have

$$n = p \times q = 5 \times 17 = 85$$

$$\phi(n) = (p-1)(q-1) = (5-1)(17-1) = 4 \times 16 = 64.$$

Find the private key d :

Given $e=13$ and $\phi(n)=64$, we need to solve:

$$13d \equiv 1 \pmod{64}$$

Use the Extended Euclidean Algorithm to find the multiplicative inverse of 13 modulo 64.

q	r_1	r_2	r	t_1	t_2	t
-	64	13	-	0	1	-
4	64	13	12	0	1	-4
1	13	12	1	1	-4	5
12	12	1	0	-4	5	-64
	1	0		5	-64	

$$5 \times 13 \equiv 1 \pmod{64}$$

So $d = 5$.

$$M = C^d \pmod{n}$$

(a) For $C = 12$,

$$M = 12^5 \pmod{85}$$

To calculate 12^5 , $12 = 4 \times 3$

$$12^5 \pmod{85} = (4 \times 3)^5 \pmod{85}$$

$$= [(4^4 \times 4 \pmod{85}) \times (3^4 \times 3 \pmod{85})] \pmod{85}$$

$$= [(256 \bmod 85 \times 4 \bmod 85) \times (81 \bmod 85 \times 3 \bmod 85)] \bmod 85$$

$$= [4 \times (-4) \times 3] \bmod 85 = -48 \bmod 85 = 37$$

$$(b) \text{ For } C = 9, M = 9^5 \pmod{85}$$

$$9^5 \bmod 85 = (9)^{2+2+1} \bmod 85$$

$$= [(81 \bmod 85) \times (81 \bmod 85) \times (9 \bmod 85)] \bmod 85$$

$$= [(-4) \times (-4) \times (9)] \bmod 85$$

$$= 16 \times 9 \bmod 85$$

$$= 144 \bmod 85$$

$$= 59$$

So for $C = 12, M = 37$;

for $C=9, M = 59$

2. Alice and Bob are using the RSA algorithm to communicate. Bob's public key is $e = 3$ and $n = 187$.

a. What is Bob's secret key?

b. Alice wants to send the message M to Bob. Bob receives 9. What was the message M sent by Alice?

Solution.

(a) $n = p \times q = 187 = 11 \times 17$, Hence $p=11, q=17$.

We calculate

$$\Phi(187) = (11-1) \times (17-1) = 10 \times 16 = 160.$$

We find d such that $3d \equiv 1 \pmod{160}$, meaning d is multiplicative inverse of 3 modulo 160.

Using the Extended Euclidean Algorithm,

q	r_1	r_2	r	t_1	t_2	t
-	160	3	-	0	1	-
53	160	3	1	0	1	-53
3	3	1	0	1	-53	160
	1	0		-53	160	

$$d = -53 \bmod 160 = 107$$

So, Bob's secret key d is 107.

(b) We have $M = C^d \pmod{n}$,

$$M = 9^{107} \pmod{187}$$

$$107 = 64 + 32 + 8 + 2 + 1 = 2^6 + 2^5 + 2^3 + 2^1 + 2^0$$

$$9^2 \pmod{187} = 81 \pmod{187}$$

$$9^4 \pmod{187} = 81^2 \pmod{187} = 6561 \pmod{187} = 16 \pmod{187}$$

$$9^8 \pmod{187} = 16^2 \pmod{187} = 256 \pmod{187} = 69 \pmod{187}$$

$$9^{16} \pmod{187} = 69^2 \pmod{187} = 4761 \pmod{187} = 86 \pmod{187}$$

$$9^{32} \pmod{187} = 86^2 \pmod{187} = 7396 \pmod{187} = 103 \pmod{187}$$

$$9^{64} \pmod{187} = 103^2 \pmod{187} = 10609 \pmod{187} = 137 \pmod{187}$$

$$M = 9^{107} \pmod{187} = 9^{64} \times 9^{32} \times 9^8 \times 9^2 \times 9 \pmod{187} = 137 \times 103 \times 69 \times 81 \times 9 \pmod{187} = 15$$

So M is 15.